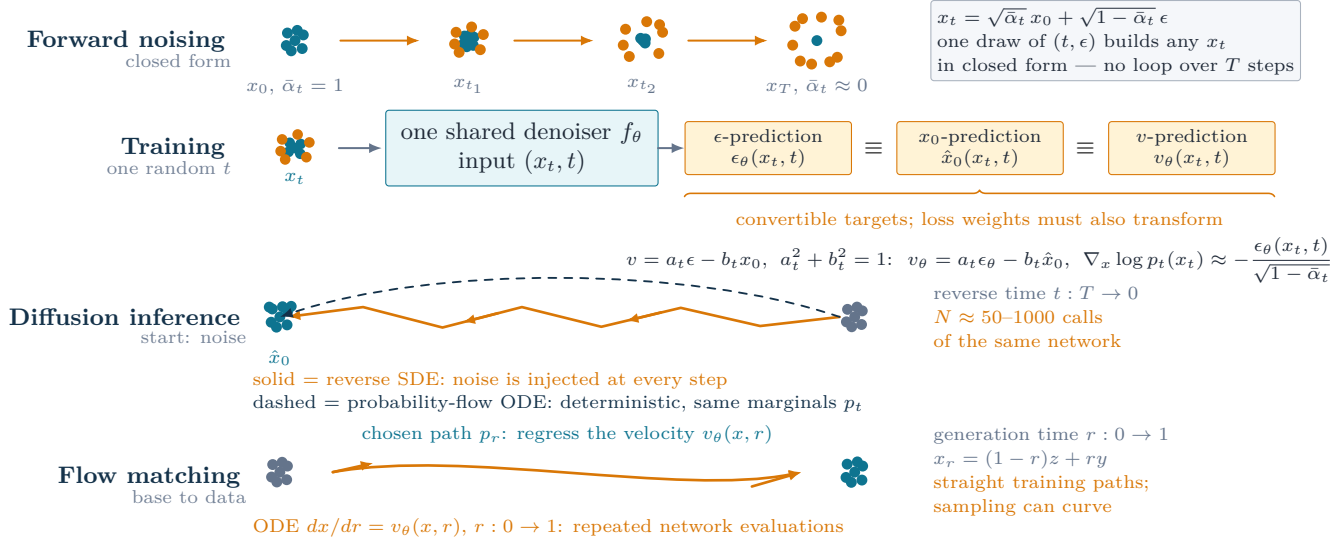
Forward noising
closed form
$x_0,\ \bar{\alpha}_t = 1$
x_{t_1}
x_{t_2}
$x_T,\ \bar{\alpha}_t \approx 0$
$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$
one draw of (t, ϵ) builds any x_t
in closed form — no loop over T steps
Training
one random t
x_t
one shared denoiser f_θ
input (x_t, t)
ϵ-prediction
$\epsilon_\theta(x_t, t)$
≡
x_0-prediction
$\hat{x}_0(x_t, t)$
≡
v-prediction
$v_\theta(x_t, t)$
convertible targets; loss weights must also transform
$v = a_t\epsilon - b_t x_0,\ a_t^2 + b_t^2 = 1$: $v_\theta = a_t\epsilon_\theta - b_t\hat{x}_0,\ \nabla_x \log p_t(x_t) \approx -\dfrac{\epsilon_\theta(x_t,t)}{\sqrt{1-\bar{\alpha}_t}}$
Diffusion inference
start: noise
$\hat{x}_0$
reverse time $t : T \to 0$
$N \approx 50$–1000 calls
of the same network
solid = reverse SDE: noise is injected at every step
dashed = probability-flow ODE: deterministic, same marginals p_t
chosen path p_r: regress the velocity $v_\theta(x, r)$
Flow matching
base to data
generation time $r : 0 \to 1$
$x_r = (1-r)z + ry$
straight training paths;
sampling can curve
ODE $dx/dr = v_\theta(x, r)$, $r : 0 \to 1$: repeated network evaluations